RV College of Engineering

Autonomous Institution Affiliated to Visvesvaraya Technological University, Belagavi | Approved by AICTE, New Delhi

Academic year 2023-2024 (Odd semester 2023)

DEPARTMENT OF MATHEMATICS

Date	20 November 2023	Time	10:00 AM to 11:30 AM
CIE I	Test I	Maximum Marks	50
Course Title	FUNDAMENTALS OF LINEAR ALGEBRA, CALCULUS AND DIFFERENTIAL EQUATIONS	Course Code	MAT211BT
Semester	I	Programs	AS, CH, IM, ME

Q. No.	Questions	M	BT	CO
1a.	Reduce the following matrix to an echelon form and hence also, find its rank. $X = \begin{bmatrix} 1 & 3 & 2 & 4 & 1 \\ 0 & 0 & 2 & 2 & 0 \\ 2 & 6 & 2 & 6 & 2 \\ 3 & 9 & 1 & 10 & 6 \end{bmatrix}$	4	2	1
1b.	Test the following equations for consistency and solve if consistent. $x_1 + x_2 - x_3 = 0,\ 2x_1 - x_2 + x_3 = 3,\ 4x_1 + 2x_2 - 2x_3 = 2.$	6	2	2
2a.	Determine the value of b such that the system of equations $2x + y + 2z = 0$, $x + y + 3z = 0,\ 4x + 3y + bz = 0$ has (i) trivial solution (ii) non-trivial solution.	4	2	1
2b.	Solve: $3x + 2y + z = 10,\ 2x + 3y + 2z = 14,\ x + 2y + 3z = 14$ by the process of Gauss-Jordan elimination.	6	3	2
3a.	Find the angle of intersection of two polar curves C_1: $r = a\theta$ and C_2: $r = \frac{a}{\theta}$, where $\theta > 0$ is the parameter.	4	1	1
3b.	Employ Rayleigh's power method to find the largest eigenvalue and the corresponding eigen vector of the matrix $A = \begin{pmatrix} 1 & -3 & 2 \\ 4 & 4 & -1 \\ 6 & 3 & 5 \end{pmatrix}$. Assume $\begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix}$ as the initial eigen vector. Perform three iterations.	6	3	3
4a.	Compute the following for the cardioid which is represented by $r = a(1 - \cos\theta)$ i) Angle between the radius vector and the tangent at any point (r, θ). ii) Angle in which the tangent to curve makes with the initial line. iii) Slope of the curve at $\theta = \pi/6$.	5	3	3
4b.	A curve that is generated by a point on the circumference of a circle as it rolls along a straight line given by the equations $x = t - \sin(t)$ and $y = 1 - \cos(t)$. Obtain the radius of curvature of the parametric curves at $t = \pi$.	5	4	3
5.	Solve the system of equations: $3x_1 + 9x_2 - 2x_3 = 11,\ 4x_1 + 2x_2 + 13x_3 = 24,\ 4x_1 - 2x_2 + x_3 = -8.$ correct up to 4 significant figures by using Gauss-Seidel method after verifying the diagonal dominance of the system. Perform 5-iterations by starting with the initial solution $x^{(0)} = y^{(0)} = z^{(0)} = 0$.	10	2	4

BT-Blooms Taxonomy, CO-Course Outcomes, M-Marks

Marks Distribution	COS / BT	CO1	CO2	CO3	CO4	L1	L2	L3	L4	L5	L6
	Max Marks	12	12	16	10	04	24	17	05	-	-

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Academic year 2023-2024 (Odd semester 2023)

DEPARTMENT OF MATHEMATICS				
Scheme and Solution				
CIE I	Test I	Date	20.11.2023	
Course Title	Fundamentals Of Linear Algebra, Calculus and Differential Equations		Course Code	MAT211BT
Semester	I	Programs	AS, CH, IM, ME	

Q. No.	Questions	M
1a	$X=\begin{bmatrix}1&3&2&4&1\\0&0&2&2&0\\2&6&2&6&2\\3&9&1&10&6\end{bmatrix}\sim\begin{bmatrix}1&3&2&4&1\\0&0&2&2&0\\0&0&-1&-2&0\\0&0&-5&-2&3\end{bmatrix}\sim\begin{bmatrix}1&3&2&4&1\\0&0&2&2&0\\0&0&0&-1&0\\0&0&0&3&3\end{bmatrix}\sim\begin{bmatrix}1&3&2&4&1\\0&0&2&2&0\\0&0&0&-1&0\\0&0&0&0&3\end{bmatrix}$ Which is the echelon form, and it contains 4 non-zero rows. **Rank of (X) = 4**	3 1
1b	$(A:B)=\left[\begin{array}{ccc:c}1&1&-1&0\\2&-1&1&3\\4&2&-2&2\end{array}\right]\sim\left[\begin{array}{ccc:c}1&1&-1&0\\0&-3&3&3\\0&-2&2&2\end{array}\right]\sim\left[\begin{array}{ccc:c}1&1&-1&0\\0&-3&3&3\\0&0&0&0\end{array}\right]$ Here, Rank (A; B) = Rank (A), System is consistent. Also Rank (A) < Number of unknowns. System has infinitely many solutions $\Rightarrow$ $x_1 = 1$, $x_2 = k-1$, $x_3 = k$.	3 3
2a	$(A:B)=\left[\begin{array}{ccc:c}2&1&2&\\1&1&3&\\4&3&b&\end{array}\right]\sim\left[\begin{array}{ccc:c}1&1&3&\\2&1&2&\\4&3&b&\end{array}\right]\sim\left[\begin{array}{ccc:c}1&1&3&\\0&-1&-4&\\0&-1&b-12&\end{array}\right]\sim\left[\begin{array}{ccc:c}1&1&3&\\0&-1&-4&\\0&0&b-8&\end{array}\right]$ (i) Trivial solution: $b \neq 8$ (ii) non-trivial solution: $b = 8$.	2 2
2b	$(A:B)=\left[\begin{array}{ccc:c}3&2&1&10\\2&3&2&14\\1&2&3&14\end{array}\right]\sim\left[\begin{array}{ccc:c}3&2&1&10\\0&\frac{5}{3}&\frac{4}{3}&\frac{22}{3}\\0&4&8&32\end{array}\right]\sim\left[\begin{array}{ccc:c}3&2&1&10\\0&\frac{5}{3}&\frac{4}{3}&\frac{22}{3}\\0&0&8&24\end{array}\right]$ $\sim\left[\begin{array}{ccc:c}3&2&0&7\\0&5/3&0&10/3\\0&0&8&24\end{array}\right]\sim\left[\begin{array}{ccc:c}3&0&0&3\\0&5/3&0&10/3\\0&0&8&24\end{array}\right]$ $\Rightarrow$ $x = 1$, $y = 2$, $z = 3$.	4 2
3a	C1: $r = a\theta \Rightarrow \frac{dr}{d\theta} = a \Rightarrow \tan(\varphi_1) = r\frac{dr}{d\theta} = (a\theta)\left(\frac{1}{a}\right) = \theta$ C2: $r = \frac{a}{\theta} \Rightarrow \frac{dr}{d\theta} = -\frac{a}{\theta^2} \Rightarrow \tan(\varphi_1) = r\frac{dr}{d\theta} = \left(\frac{a}{\theta}\right)\left(-\frac{\theta^2}{a}\right) = -\theta$ $\lvert\varphi_1 - \varphi_2\rvert = tan^{-1}\left(\frac{2\theta}{1-\theta^2}\right)$	3 1
3b	$X_1 = AX_0 = \begin{pmatrix}1&-3&2\\4&4&-1\\6&3&5\end{pmatrix}\begin{pmatrix}1\\0\\0\end{pmatrix} = \begin{pmatrix}1\\4\\6\end{pmatrix} = 6\begin{pmatrix}0.1666\\0.6666\\1\end{pmatrix} = \lambda^{(1)}X_1$	2

	$X_2 = AX_1 = \begin{pmatrix}1 & -3 & 2\\ 4 & 4 & -1\\ 6 & 3 & 5\end{pmatrix}\begin{pmatrix}0.1666\\ 0.6666\\ 1\end{pmatrix} = \begin{pmatrix}0.1666\\ 2.3333\\ 8\end{pmatrix} = 8\begin{pmatrix}0.2083\\ 0.2916\\ 1\end{pmatrix} = \lambda^{(2)} X_2$ $X_3 = AX_2 = \begin{pmatrix}1 & -3 & 2\\ 4 & 4 & -1\\ 6 & 3 & 5\end{pmatrix}\begin{pmatrix}0.2083\\ 0.2916\\ 1\end{pmatrix} = \begin{pmatrix}1.1458\\ 0.2500\\ 6\end{pmatrix} = 6\begin{pmatrix}0.1909\\ 0.0416\\ 1\end{pmatrix} = \lambda^{(3)} X_3$ Largest eigenvalue = 6 and Corresponding eigenvector = $\begin{pmatrix}0.1909\\ 0.0416\\ 1\end{pmatrix}$	**2** **2**
4a	$r = a(1-\cos\theta) \Rightarrow \frac{dr}{d\theta} = a\sin\theta \Rightarrow \tan(\varphi) = r\frac{dr}{d\theta} = \frac{a(1-\cos\theta)}{a\sin\theta} = \tan(\theta/2)$ i) Angle between the radius vector and the tangent = $\varphi = (\theta/2)$ ii) Angle in which the curve makes with the initial line = $\psi = \theta + \varphi = 3\theta/2$. iii) Slope of the curve at $\theta = \pi/6$: $\tan(\psi) = \tan(\theta+\varphi) = \tan(3\theta/2) = \tan(\pi/4) = 1$.	**2** **3**
4b	$x = t - \sin(t) \Rightarrow \dot{x} = 1-\cos(t) \Rightarrow \ddot{x} = \sin(t)$ $y = 1-\cos(t) \Rightarrow \dot{y} = \sin(t) \Rightarrow \ddot{y} = \cos(t)$ $\rho = \frac{(\dot{x}^2+\dot{y}^2)^{\frac{3}{2}}}{\dot{x}\ddot{y}-\dot{y}\ddot{x}} = \frac{((1-\cos t)^2+(\sin t)^2)^{\frac{3}{2}}}{\cos(t)-\cos^2(t)-\sin^2(t)} = -2$ $\Rightarrow \lvert\rho\rvert = 2$ and $\rho_{(t=\pi)} = 2$.	**2** **2** **1**
5	Gauss-Seidel iterative formula after rearranging into diagonally dominant system: $x_1^{(k+1)} = \left(\frac{1}{4}\right)\left(-8 + 2x_2^{(k)} - x_3^{(k)}\right)$ $x_2^{(k+1)} = \left(\frac{1}{9}\right)\left(11 - 3x_1^{(k+1)} + 2x_3^{(k)}\right)$ $x_3^{(k+1)} = \left(\frac{1}{13}\right)\left(24 - 4x_1^{(k+1)} - 2x_2^{(k+1)}\right)$ k = 0, 1, 2, 3, 4 Initial solution: $x_1^{(0)} = x_2^{(0)} = x_3^{(0)} = 0$ **NOTE: Alternative approach to the solution may be awarded full marks**	**2** **8**

Iterations	x_1	x_2	x_3
1	-2	1.8889	2.1709
2	-1.5983	2.2374	1.9937
3	-1.3797	2.1252	1.9437
4	-1.4233	2.1286	1.9566
5	-1.4249	2.1320	1.9566

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Academic year 2023-2024 (Odd semester 2023)

DEPARTMENT OF MATHEMATICS				
Date	**27 December 2023**	Time	**10:00 AM to 11:30 AM**	
CIE II	**Test II**	Maximum Marks	**50**	
Course Title	**FUNDAMENTALS OF LINEAR ALGEBRA, CALCULUS AND DIFFERENTIAL EQUATIONS**		Course Code	**MAT211BT**
Semester	**I**	Programs	**AS, CH, IM, ME, SPARK**	

Q. No.	Questions	M	BT	CO
1a.	Show that the radius of curvature of the curve $r^n = a^n cosn\theta$ is $\frac{a^n r^{1-n}}{(n+1)}$ for $n \neq -1$.	5	2	1
1b.	Find the centre of curvature of the parabola $y^2 = 4ax$ at the point $(a, 2a)$.	5	2	2
2a.	Using Maclaurin's series expand e^{sinx} up to the terms containing 4th degree.	5	3	2
2b.	If $u = xlog(xy)$ where $x^3 + y^3 + 3xy = 1$ then find $\frac{du}{dx}$.	5	1	1
3a.	If $u = f(2x - 3y,\ 3y - 4z, 4z - 2x)$, then prove that $\frac{1}{2}\frac{\partial u}{\partial x} + \frac{1}{3}\frac{\partial u}{\partial y} + \frac{1}{4}\frac{\partial u}{\partial z} = 0$.	5	3	3
3b.	Find $\frac{du}{dt}$ when $u = sin\left(\frac{x}{y}\right)$, $x = e^t$, $y = t^2$.	5	3	3
4.	A rectangular box, which is open at the top, has a capacity of 256 cubic feet. Determine the dimensions of the box such that the least material is required for the construction of the box. Use Lagranges's method of multipliers to obtain the solution.	10	4	3
5.	If $x = \sqrt{vw}, y = \sqrt{wu}, z = \sqrt{uv}$ and $u = r\ sin\theta\ cos\phi, v = r\ sin\theta\ sin\phi$, $w = r\ cos\theta$. Calculate $\frac{\partial(x,y,z)}{\partial(r,\theta,\phi)}$.	10	2	4

BT-Blooms Taxonomy, CO-Course Outcomes, M-Marks

Marks Distribution	COS / BT	CO1	CO2	CO3	CO4	L1	L2	L3	L4	L5	L6
	Max Marks	10	10	20	10	05	20	15	10	-	-

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RV Educational Institutions
RV College of Engineering

Autonomous Institution Affiliated to Visvesvaraya Technological University, Belagavi

Approved by AICTE, New Delhi

DEPARTMENT OF MATHEMATICS
Scheme and Solution – Physics cycle

Course: Fundamentals of Linear Algebra, Calculus and Differential Equations.	**CIE – 2 (TEST)**	**Maximum marks: 50**
Course code:MAT211BT	**First Semester 2023-2024 Branches: AS, CH, IEM, ME, SPARK(PH)**	

Q.No		Marks
1 a)	$r^n = a^n cosn\theta$ -----(1). Radius of curvature in polar form $\rho = \frac{\left(r^2+\left(\frac{dr}{d\theta}\right)^2\right)^{3/2}}{r^2+2\left(\frac{dr}{d\theta}\right)^2 - r\frac{d^2r}{d\theta^2}}$.	
	On differentiating (1) w. r. t θ	
	$\frac{dr}{d\theta} = \frac{-a^n sinn\theta}{r^{n-1}} = -rtann\theta$, $\frac{d^2r}{d\theta^2} = rtan^2n\theta - nrsec^2n\theta$	1+2
	Substituting $\frac{dr}{d\theta}, \frac{d^2r}{d\theta^2}$ in radius of curvature equation we get, $\rho = \frac{a^n r^{1-n}}{n+1}$	2
1 b)	$y^2 = 4ax$--------(1). On differentiating (1) w. r. t x	
	$y_1 at\ (a, 2a) = 1$, $y_2 at\ (a, 2a) = -\frac{1}{2a}$ substituting in $\bar{x}$ and $\bar{y}$ equation	1+1
	in $\bar{x} = x - \frac{y_1(1+y_1^2)}{y_2} = 5a$, $\bar{y} = y + \frac{(1+y_1^2)}{y_2} = -2a$	1+1
	Centre of curvature = $(\bar{x}, \bar{y}) = (5a, -2a)$.	1
2 a)	$y = e^{sinx}$---(1). Maclaurin's series of (1) is $y = y(0) + \frac{xy'(0)}{1!} + \frac{x^2y''(0)}{2!} + \frac{x^3y'''(0)}{3!} + \frac{x^4y^{iv}(0)}{4!} + \cdots$	
	Derivatives of (1) are $y(0) = 1$, $y'(0) = 1$, $y''(0) = 1$, $y'''(0) = 0$, $y^{iv}(0) = -3$	4
	$y = 1 + x + \frac{x^2}{2} - \frac{x^4}{8}$	1
2 b)	$u = xlog(xy)$. $\frac{\partial u}{\partial x} = 1 + \log(xy)$, $\frac{\partial u}{\partial y} = \frac{x}{y}$	1+1
	From $x^3 + y^3 + 3xy = 1$, we get $\frac{dy}{dx} = -\left(\frac{x^2+y}{y^2+x}\right)$	
	Total differential w. r. t 'x' $\frac{du}{dx} = \frac{\partial u}{\partial x} + \frac{\partial u}{\partial y}\frac{dy}{dx} = (1 + \log(xy)) - \frac{x}{y}\left(\frac{x^2+y}{y^2+x}\right) = \frac{y^3 - x^3 + log xy(y^3+xy)}{y(y^2+x)}$	3

3 (a)	Let u=u(L, M, N). Where L=2x - 3y, M = 3y - 4z, N = 4z - 2x By Chain Rule $\frac{\partial u}{\partial x} = \frac{\partial u}{\partial L}\cdot\frac{\partial L}{\partial x} + \frac{\partial u}{\partial M}\cdot\frac{\partial M}{\partial x} + \frac{\partial u}{\partial N}\cdot\frac{\partial N}{\partial x} = \frac{\partial u}{\partial L}(2) + \frac{\partial u}{\partial M}(0) + \frac{\partial u}{\partial N}(-2)$-------(1) $\frac{\partial u}{\partial y} = \frac{\partial u}{\partial L}\cdot\frac{\partial L}{\partial y} + \frac{\partial u}{\partial M}\cdot\frac{\partial M}{\partial y} + \frac{\partial u}{\partial N}\cdot\frac{\partial N}{\partial y} = \frac{\partial u}{\partial L}(-3) + \frac{\partial u}{\partial M}(3) + \frac{\partial u}{\partial N}(0)$-------(2) $\frac{\partial u}{\partial z} = \frac{\partial u}{\partial L}\cdot\frac{\partial L}{\partial z} + \frac{\partial u}{\partial M}\cdot\frac{\partial M}{\partial z} + \frac{\partial u}{\partial N}\cdot\frac{\partial N}{\partial z} = \frac{\partial u}{\partial L}(0) + \frac{\partial u}{\partial M}(-4) + \frac{\partial u}{\partial N}(4)$-------(3) $\therefore \frac{1}{2}\frac{\partial u}{\partial x} + \frac{1}{3}\frac{\partial u}{\partial y} + \frac{1}{4}\frac{\partial u}{\partial z} = 0$	4 1
3 (b)	$\frac{du}{dt} = \frac{\partial u}{\partial x}\frac{dx}{dt} + \frac{\partial u}{\partial y}\frac{dy}{dt}$ $= cos\left(\frac{x}{y}\right)\frac{1}{y}e^t + cos\left(\frac{x}{y}\right)\left(-\frac{x}{y^2}\right)2t$ $= cos\left(\frac{e^t}{t^2}\right)e^t\left(\frac{t-2}{t^3}\right)$.	1 4
4	Let x, y, z be the length, breadth and height of the box. Volume = $xyz = 256 \Rightarrow xyz - 256 = 0$. $\phi(x,y,z) = xyz - 256$. Let S be the material surface of the box. $S = xy + 2yz + 2zx$. $\frac{\partial S}{\partial x} = y + 2z, \frac{\partial S}{\partial y} = x + 2z, \frac{\partial S}{\partial z} = 2y + 2x$ and $\frac{\partial \phi}{\partial x} = yz, \frac{\partial \phi}{\partial y} = xz, \frac{\partial \phi}{\partial z} = xy$ Using the Lagrange's method of multiplier, we get $y + 2z + \lambda yz = 0, \quad x + 2z + \lambda xz = 0, \quad 2y + 2x + \lambda xy = 0$ Multiply the above equations respectively by x, y and z and put $xyz = 256$, we get $xy + 2xz = -256\lambda, \quad xy + 2yz = -256\lambda, \quad 2yz + 2zx = -256\lambda$ Solving above yields $x = y = 2z$. Put this value in $xyz = 256$ to get $x = y = 8, z = 4$.	1 2 2 1 2 2
5	If $x = \sqrt{vw}, y = \sqrt{wu}, z = \sqrt{uv}$ and $u = r\,sin\theta\,cos\phi, v = r\,sin\theta\,sin\phi, w = r\,cos\theta$. $\frac{\partial(x,y,z)}{\partial(u,v,w)} = \begin{vmatrix} 0 & \frac{\sqrt{w}}{2\sqrt{v}} & \frac{\sqrt{v}}{2\sqrt{w}} \\ \frac{\sqrt{w}}{2\sqrt{u}} & 0 & \frac{\sqrt{u}}{2\sqrt{w}} \\ \frac{\sqrt{v}}{2\sqrt{u}} & \frac{\sqrt{u}}{2\sqrt{v}} & 0 \end{vmatrix} = \frac{1}{4'}\frac{\partial(u,v,w)}{\partial(r,\theta,\phi)} = \begin{vmatrix} \sin\theta\,cos\phi & r\,cos\theta\,cos\phi & -r\,sin\theta\,sin\phi \\ sin\theta\,sin\phi & r\,cos\theta\,sin\phi & r\,sin\theta\,cos\phi \\ cos\theta & -r\,sin\theta & 0 \end{vmatrix} = r^2 sin\theta$ $\frac{\partial(x,y,z)}{\partial(r,\theta,\phi)} = \frac{1}{4}r^2 sin\theta$.	5+4+1

Note: Alternate correct methods may be awarded appropriate marks.

Academic year 2023-2024 (Odd semester)

DEPARTMENT OF MATHEMATICS				
Date	**22nd January 2024**	Time	**10:00 AM to 11:30 AM**	
CIE I	**Improvement Test**	Maximum Marks	**50**	
Course Title	**FUNDAMENTALS OF LINEAR ALGEBRA, CALCULUS AND DIFFERENTIAL EQUATIONS**		Course Code	**MA211TB**
Semester	**I**	Programs	**AS, CH, IM, ME**	

Q. No.	Questions	M	BTL	CO
1a	If $x = e^{r\cos(\theta)}\cos(r\sin(\theta))$ and $y = e^{r\cos(\theta)}\sin(r\sin(\theta))$, then prove that. $$\frac{\partial^2 x}{\partial\theta^2} + r\frac{\partial x}{\partial r} + r^2\frac{\partial^2 x}{\partial r^2} = 0$$	5	1	1
1b	If $z = \sqrt{x^2+y^2}$ and $x^3+y^3+3axy = 5a^2$ find the value of $\frac{dz}{dx}$, when $x = y = a$.	5	2	1
2	The temperature T at any point (x, y, z) in space is given by $T = 400xyz^2$, find the highest temperature on the surface of the unit sphere $x^2+y^2+z^2 = 1$.	10	4	4
3a	Solve the initial value problem $2x''(t) + 5x'(t) - 3x(t) = 0, x(0) = 1, x'(0) = 4$.	5	2	2
3b	Find the general solution of $y'' - 4y = xe^x$.	5	2	2
4	Using Variation of parameters, solve $\frac{d^2y}{dx^2} - y = \frac{2}{1+e^x}$	10	3	3
5	Solve $x^2\frac{d^2y}{dx^2} - 3x\frac{dy}{dx} + 5y = x^2\sin(\log_e(x))$.	10	3	3

BT-Blooms Taxonomy, CO-Course Outcomes, M-Marks

Marks Distribution	COS / BT	CO1	CO2	CO3	CO4	L1	L2	L3	L4	L5	L6
	Max Marks	10	10	20	10	5	15	20	10	-	-

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Academic year 2023-2024 (Odd semester)

DEPARTMENT OF MATHEMATICS				
Date	22nd January 2024	Time	10:00 AM to 11:30 AM	
CIE I	Improvement Test Scheme and solution	Maximum Marks	50	
Course Title	FUNDAMENTALS OF LINEAR ALGEBRA, CALCULUS AND DIFFERENTIAL EQUATIONS		Course Code	MA211TB
Semester	I	Programs	AS, CH, IM, ME	

Q. No.	Questions	M
1a	If $x = e^{r\cos(\theta)}\cos(r\sin(\theta))$ and $y = e^{r\cos(\theta)}\sin(r\sin(\theta))$	
	$\frac{\partial x}{\partial \theta} = -re^{r\cos(\theta)}\sin(\theta + r\sin(\theta))$ $\frac{\partial x}{\partial r} = e^{r\cos(\theta)}\cos(\theta + r\sin(\theta))$	1
	$\frac{\partial y}{\partial \theta} = re^{r\cos(\theta)}\cos(\theta + r\sin(\theta))$ $\frac{\partial y}{\partial r} = e^{r\cos(\theta)}\sin(\theta + r\sin(\theta))$	1
	$\frac{\partial x}{\partial \theta} = -r\frac{\partial y}{\partial r} \Rightarrow \frac{\partial^2 x}{\partial \theta^2} = -r\frac{\partial^2 y}{\partial \theta\,\partial r} = -r\frac{\partial^2 y}{\partial r\,\partial \theta}$, $\frac{\partial x}{\partial r} = \frac{1}{r}\frac{\partial y}{\partial \theta} \Rightarrow \frac{\partial^2 x}{\partial r^2} = \frac{1}{r^2}\frac{\partial y}{\partial \theta} + \frac{1}{r}\frac{\partial y}{\partial r\,\partial \theta}$	2
	$\frac{\partial^2 x}{\partial \theta^2} + r\frac{\partial x}{\partial r} + r^2\frac{\partial^2 x}{\partial r^2} = -r\frac{\partial^2 y}{\partial r\,\partial \theta} + \frac{\partial y}{\partial \theta} - \frac{\partial y}{\partial \theta} + r\frac{\partial^2 y}{\partial r\,\partial \theta} = 0$	1
1b	If $z = \sqrt{x^2+y^2}$ and $x^3 + y^3 + 3axy = 5a^2$ then $\frac{dz}{dx} = \frac{\partial z}{\partial x} + \frac{\partial z}{\partial y}\frac{dy}{dx}$	1
	$\frac{\partial z}{\partial x} = \frac{x}{\sqrt{x^2+y^2}}$, $\frac{\partial z}{\partial x} = \frac{y}{\sqrt{x^2+y^2}}$, $\frac{dy}{dx} = -\frac{x^2+ay}{y^2+ax} \Rightarrow \frac{dz}{dx} = \frac{x}{\sqrt{x^2+y^2}} + \frac{y}{\sqrt{x^2+y^2}}\left(-\frac{x^2+ay}{y^2+ax}\right)$	3
	At $x = y = a$ $\frac{dz}{dx} = \frac{a}{\sqrt{a^2+a^2}} + \frac{a}{\sqrt{a^2+a^2}}\left(-\frac{a^2+aa}{a^2+aa}\right) = 0$	1
2	$F = 400\,xyz^2 + \lambda(x^2+y^2+z^2-1)$	1
	$F_x = 0 \Rightarrow T + 2\lambda x^2 = 0$, $F_y = 0 \Rightarrow T + 2\lambda y^2 = 0$ and $F_z = 0 \Rightarrow 2T + 2\lambda z^2 = 0$	3
	Adding above equations and from $x^2+y^2+z^2 = 1$ we get, $4T + 2\lambda = 0 \Rightarrow \lambda = -2T$	2
	Therefore $x = y = z = \pm\frac{1}{2}$ hence $T = 400xyz^2 = \frac{400}{8} = 50$	3 1
3a	$2x''(t) + 5x'(t) - 3x(t) = 0$, $2m^2 + 5m - 3 = 0 \Rightarrow m = \frac{1}{2}, -3 \Rightarrow x(t) = c_1 e^{\frac{t}{2}} + c_2 e^{-3t}$	2
	When $x(0) = 1$, $x'(0) = 4. \Rightarrow 1 = c_1 + c_2$ and $8 = c_1 - 6c_2 \Rightarrow c_1 = 2$ and $c_2 = -1$	2
	$x(t) = 2e^{\frac{t}{2}} - e^{-3t}$	1
3b	$y'' - 4y = xe^x \Rightarrow$ AE: $m^2 - 4 = 0 \Rightarrow m = \pm 2 \Rightarrow CF = c_1e^{2x} + c_2e^{-2x}$	1
	PI: $\frac{1}{D^2-1}xe^x \Rightarrow e^x\frac{1}{(D+1)^2-1}x \Rightarrow e^x\frac{1}{D^2+2D+1-1}x \Rightarrow e^x\frac{1}{2D}\frac{1}{\left(1+\frac{D}{2}\right)}x \Rightarrow e^x\frac{1}{2D}\left(1-\frac{D}{2}+\frac{D^2}{2}\ldots\right)x$	2
	$e^x\frac{1}{2D}\left(x - \frac{Dx}{2} + \frac{D^2x}{2}\ldots\right) \Rightarrow e^x\frac{1}{2D}\left(x-\frac{1}{2}\right) \Rightarrow \frac{e^x}{2}\int\left(x-\frac{1}{2}\right)dx \Rightarrow \frac{e^x}{2}\left(\frac{x^2}{2} - \frac{x}{2}\right)$	2

Academic year 2023-2024 (Odd semester)

4

$\frac{d^2y}{dx^2} - y = \frac{2}{1+e^x} \Rightarrow (D^2-1)y = \frac{2}{1+e^x} \Rightarrow m^2 - 1 = 0 \Rightarrow m = \pm 1 \quad CF = c_1e^x + c_2e^{-x}$ — 2

$y_1 = e^x \quad y_2 = e^{-x} \quad w = -2$ — 2

$A' = -\frac{y_2\ \phi(x)}{w} = -\frac{e^{-x}\frac{2}{1+e^x}}{-2} = \frac{1}{e^x(1+e^x)} \Rightarrow A = \int \frac{1}{e^x(1+e^x)}\,dx$

Put $e^x = t \Rightarrow e^x dx = dt \Rightarrow dx = \frac{dt}{t} \Rightarrow A = \int \frac{1}{t^2(1+t)}\,dt \Rightarrow A = \int \left(\frac{1}{t^2} - \frac{1}{t} + \frac{1}{1+t}\right) dt$

$A = -\frac{1}{t} - \log t + \log(1+t) \Rightarrow A = -e^{-x} - x + \log(1+e^x)$ — 3

$B' = \frac{y_1\ \phi(x)}{w} = \frac{e^x\frac{2}{1+e^x}}{-2} = -\frac{e^x}{(1+e^x)} \Rightarrow B = \int -\frac{e^x}{(1+e^x)}\,dx = -\log(1+e^x)$ — 2

$PI = Ay_1 + By_2 = e^x(-e^{-x} - x + \log(1+e^x)) - e^{-x}\log(1+e^x)$

$y(t) = CF + PI$ — 1

5

$x^2\frac{d^2y}{dx^2} - 3x\frac{dy}{dx} + 5y = x^2\sin(\log_e(x))$. — 1

Let $x = e^t \Rightarrow t = \ln x$

$(D(D-1) - 3D + 5) = e^{2t}\sin(t) \Rightarrow D^2 - 4D + 5 = e^{2t}\sin(t)$ — 1

AE: $m^2 - 4m + 5 = 0 \Rightarrow m = 2 \pm i$ — 2

CF: $e^{2t}(c_1\cos(t) + c_2\sin(t)) \Rightarrow x^2(c_1\cos(\ln(x)) + c_2\sin(\ln(x)))$ — 1

PI: $\frac{1}{D^2-4D+5}\,e^{2t}\sin(t) = e^{2t}\frac{1}{(D+2)^2-4(D+2)+5}\sin(t) = e^{2t}\frac{1}{D^2+1}\sin(t) = t\,e^{2t}\frac{1}{2D}\sin(t)$ — 2

$= \frac{te^{2t}}{2}\int \sin(t)\,dt = -\frac{te^{2t}}{2}\cos(t) \Rightarrow PI = -\frac{x^2\ln(x)}{2}\cos(\log x)$ — 1

$y(x) = CF + PI$ — 1

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RV COLLEGE OF ENGINEERING®

(An Autonomous Institution Affiliated to VTU)

I Semester B. E. Examinations May-2023

(Common to AS, CH, IM, ME)

FUNDAMENTALS OF LINEAR ALGEBRA, CALCULUS AND DIFFERENTIAL EQUATIONS

Time: 03 Hours *Maximum Marks: 100*

Instructions to candidates:

1. Answer all questions from Part A. Part A questions should be answered in first three pages of the answer book only.
2. Answer FIVE full questions from Part B. In Part B question number 2 is compulsory. Answer any one full question from 3 and 4, 5 and 6, 7 and 8, 9 and 10.
3. Use of scientific calculator is permitted.
4. Use of mathematics Handbook is permitted. Do not write anything on handbook.

PART-A

1	1.1	The rank of the matrix $A = \begin{bmatrix} 1 & a & a^2 \\ 1 & b & b^2 \\ 1 & c & c^2 \end{bmatrix}$, where $a = b \neq c$ is _____.	02
	1.2	If two of the eigenvalues of $A = \begin{bmatrix} 3 & -1 & 1 \\ -1 & 5 & -1 \\ 1 & -1 & 3 \end{bmatrix}$ are 3 and 6, then the eigenvalues of A^{-1} are _____.	02
	1.3	The circle $x^2 + y^2 - 8x = 0$ in polar form is _____.	02
	1.4	The coordinates of the center of curvature on the curvey $= e^x$ at a point where curve crosses y-axis are _____.	01
	1.5	The radius of curvature at the point $(-1,1)$ on the curve $y = x^2 - 3x + 1$, given $y_1 = -1$ and $y_2 = 2$ is _____.	02
	1.6	If $u = x^y$ then the first order partial derivatives are $\frac{\partial u}{\partial x}$ = _____ and $\frac{\partial u}{\partial y}$ = _____.	01
	1.7	Given $f_{xx} = 6x, f_{xy} = 12$ and $f_{yy} = 6y$ then the nature of stationary point at $(-1,5)$ is _____.	02
	1.8	Transform the integral $\int_{\theta=0}^{\pi/2} \int_{r=0}^{\infty} e^{-r^2} r \, dr \, d\theta$ to cartesian form.	02
	1.9	Sketch the region of the integration $\int_0^1 \int_{e^x}^{e} \frac{1}{\log_e y} dy \, dx$	02
	1.10	If 3 and $-\frac{1}{2}$ are the roots of an auxiliary equation, then the corresponding differential equation is _____.	02
	1.11	The particular integral of the differential equation $y'' - 4y = 3^x$ is _____.	02

PART-B

2	a	Test for consistency of the given system of equations and solve by Gauss elimination method. $x + y + z = 6$ $x - y + 2z = 5$ $3x + y + z = 8$	06
	b	The temperatures x_1, x_2, x_3 of a metal plate follow the linear equations, $2x_1 + 8x_2 - x_3 = 11$ $5x_1 - x_2 + x_3 = 10$ $-x_1 + x_2 + 4x_3 = 3.$ Compute x_1, x_2 and x_3 using Gauss-Seidel iteration method. Perform three iterations.	05

	c	Perform five iterations of the Rayleigh's power method to determine the dominant eigenvalue and eigenvector of the matrix. Use $X_0 = [1,1,1]^T$ as the initial approximation. $A = \begin{bmatrix} 6 & -2 & 2 \\ -2 & 3 & -1 \\ 2 & -1 & 3 \end{bmatrix}$	05
3	a	Determine the angle of intersection of the curves $r = a\log_e\theta$ and $r = \frac{a}{\log_e\theta}$.	08
	b	Determine the radius of curvature of the curve $x = a\left(cost + \log_e\left(tan\frac{t}{2}\right)\right), y = asin\,t$.	08
		OR	
4	a	Obtain the angle between the radius vector and tangent at any point on the curve $r = a(1 + sin\theta)$.	08
	b	Expand $\log_e(1 + e^x)$ in ascending powers of 'x' up to the term containing x^4.	08
5	a	If $\theta = t^n e^{-\frac{r^2}{4t}}$, find the value of n such that $\frac{1}{r^2}\frac{\partial}{\partial r}\left(r^2\frac{\partial\theta}{\partial r}\right) = \frac{\partial\theta}{\partial t}$.	08
	b	Determine the points on the sphere $x^2 + y^2 + z^2 = 4$ that are closest to and farthest from the point $(3,1,-1)$.	08
		OR	
6	a	Prove that $y = f(x + at) + g(x - at)$ is the solution of one-dimensional wave equation $\frac{\partial^2 y}{\partial t^2} = a^2\frac{\partial^2 y}{\partial x^2}$, where a is a constant.	08
	b	A rectangular box open at the top is to have a volume of 32 cubic ft. Determine the dimensions of the box requiring least material for its construction.	08
7	a	Evaluate the given integral by changing the order of integration $\int_0^1\int_{\sqrt{x}}^1\sqrt{1+y^3}\,dy\,dx$.	08
	b	Evaluate the given integral by changing the variables to polar coordinates. $\int_{-1}^1\int_{-\sqrt{1-x^2}}^0 \cos(x^2+y^2)\,dy\,dx$.	08
		OR	
8	a	Find the area lying inside the circle $r = asin\theta$ and outside the cardioid $r = a(1 + cos\theta)$.	08
	b	Determine the mass of the tetrahedron bounded by the coordinate planes and the plane $\frac{x}{a} + \frac{y}{b} + \frac{z}{c} = 1$. The variable density $\rho = \mu xyz$, where μ is a constant.	08
9	a	Obtain the general solution of the differential equation $y'' - 6y' + 9y = e^{3x}(x^2 + 7x + 5)$.	08
	b	Solve the following differential equation by the method of variation of parameters $\frac{d^2y}{dx^2} + 16y = 32\sec 2x$.	08
		OR	
10	a	Solve the differential equation $x^2\frac{d^2y}{dx^2} - x\frac{dy}{dx} + 2y = xlog_e x$.	08
	b	The displacement $x(t)$ of a mass-spring system is governed by $\frac{d^2x}{dt^2} + \lambda\frac{dx}{dt} + kx = f(t)$, where λ and k are force constants, determine $x = x(t)$, under the conditions $x(0) = 0 = x'(0)$, and $\lambda = 4$, $k = 20$, $f(t) = 10cos2t$.	08

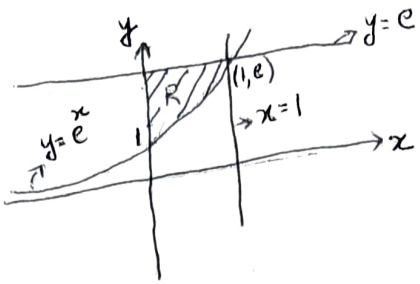
RV Educational Institutions
RV College of Engineering
Autonomous Institution Affiliated to Visvesvaraya Technological University, Belagavi
Approved by AICTE, New Delhi

DEPARTMENT OF MATHEMATICS

Scheme and Solution

Course: Fundamentals of Linear Algebra, Calculus and Differential Equations	**BE Exam April 2023**	**Maximum marks: 100**
Course code:22MA11B	**First semester 2022-2023 Physics Cycle Branches: AS, CH, IM, ME-A, ME-B, SPARK(PH)**	

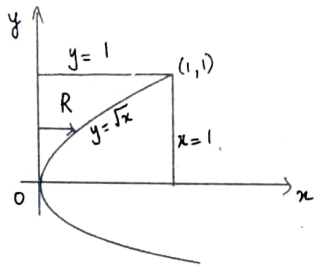
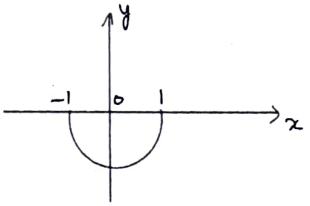
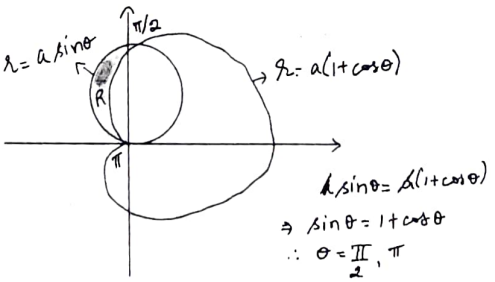
Q.No	**PART A**	Marks
1.1	Since $a = b \neq c$ $A = \begin{bmatrix} 1 & a & a^2 \\ 1 & a & a^2 \\ 1 & c & c^2 \end{bmatrix}$ $R_2 \to R_2 - R_1$ $A \sim \begin{bmatrix} 1 & a & a^2 \\ 0 & 0 & 0 \\ 1 & c & c^2 \end{bmatrix}$ $R_3 \leftrightarrow R_2$ $A \sim \begin{bmatrix} 1 & a & a^2 \\ 1 & c & c^2 \\ 0 & 0 & 0 \end{bmatrix}$ Rank of the matrix is 2.	1 1
1.2	$\lambda_1 + \lambda_2 + \lambda_3 = 3 + 5 + 3 = 11$ $9 + \lambda_3 = 11 \Rightarrow \lambda_3 = 2$ The eigenvalues of A^{-1} are $\frac{1}{3}, \frac{1}{6}, \frac{1}{2}$	1 1
1.3	$r = 8\cos\theta$	2
1.4	$\bar{x} = x - \frac{y_1(1+y_1^2)}{y_2} = 0 - \frac{1(1+1)}{1} = -2$ $\bar{y} = y + \frac{(1+y_1^2)}{y_2} = 1 + \frac{(1+1)}{1} = 3$	1
1.5	$\rho = \frac{(1+y_1^2)^{\frac{3}{2}}}{y_2} = \frac{(1+1)^{\frac{3}{2}}}{2} = \frac{2\sqrt{2}}{2} = \sqrt{2}$	2
1.6	$\frac{\partial u}{\partial x} = yx^{y-1}, \frac{\partial u}{\partial y} = x^y \log_e x$	1
1.7	At $(-1,5)$, $f_{xx} = -6, f_{xy} = 12, f_{yy} = 30 \Rightarrow f_{xx} \cdot f_{yy} - f_{xy}^2 = -180 - 144 = -324 < 0$ $\therefore (-1,5)$ is a saddle point.	1 1
1.8	$\int_{\theta=0}^{\pi/2} \int_{r=0}^{\infty} e^{-r^2} r \, dr \, d\theta = \int_{y=0}^{\infty} \int_{x=0}^{\infty} e^{-(x^2+y^2)} \, dx \, dy$	2

1.9		2
1.10	$(m-3)\left(m+\frac{1}{2}\right)=0 \Rightarrow m^2-\frac{5}{2}m-\frac{3}{2}=0 \Rightarrow 2m^2-5m-3=0$ $2\frac{d^2y}{dx^2}-5\frac{dy}{dx}-3y=0.$	1 1
1.11	$y_p=\frac{1}{D^2-4}3^x=\frac{1}{D^2-4}e^{x\ln 3}=\frac{1}{[(\ln 3)^2-4]}3^x$	2

Q.No	PART B	Marks
2a)	$[A:B]=\left(\begin{array}{ccc\|c}1&1&1&6\\1&-1&2&5\\3&1&1&8\end{array}\right)$ $R_2 \to R_2-R_1, R_3 \to R_3-3R_1$ $[A:B]=\left(\begin{array}{ccc\|c}1&1&1&6\\0&-2&1&-1\\0&-2&-2&-10\end{array}\right)$ $R_3 \to R_3-R_2$ $[A:B]=\left(\begin{array}{ccc\|c}1&1&1&6\\0&-2&1&-1\\0&0&-3&-9\end{array}\right)----(1)$ $\therefore rank(A)=rank([A:B])$=no. of unknowns=3 Hence the given system of equations is consistent and will have unique solution. (1) can be written as $x+y+z=6$ $-2y+z=-1$ $-3z=-9 \Rightarrow z=3$ Solving the above system we get $x=1, y=2, z=3$	1 1 1 3
2b)	The given system is not strictly diagonally dominant. So we interchange the first and second equations $5x_1-x_2+x_3=10$ $2x_1+8x_2-x_3=11$ $-x_1+x_2+4x_3=3$ $x_1^k=\frac{1}{5}[10+x_2^{k-1}-x_3^{k-1}]$ $x_2^k=\frac{1}{8}[11-2x_1^k+x_3^{k-1}]$ $x_3^k=\frac{1}{4}[3+x_1^k-x_2^k]$ Let $x_1^0=0, x_2^0=0, x_3^0=0$ be the initial approximation. 1st iteration: $k=1 \Rightarrow x_1^1=2, x_2^1=0.875, x_3^1=1.0313$	1 1

	2nd iteration: $k = 2 \Rightarrow x_1^2 = 1.9687, x_2^2 = 1.0117, x_3^2 = 0.9893$ 3rd iteration: $k = 3 \Rightarrow x_1^3 = 2.0045, x_2^3 = 0.9975, x_3^3 = 1.0018$	3
2c)	$$AX_0 = \begin{bmatrix}6\\0\\4\end{bmatrix} = 6\begin{bmatrix}1\\0\\0.67\end{bmatrix} = \lambda_1 X_1$$ $$AX_1 = \begin{bmatrix}7.34\\-2.67\\4.01\end{bmatrix} = 7.34\begin{bmatrix}1\\-0.36\\0.55\end{bmatrix} = \lambda_2 X_2$$ $$AX_2 = \begin{bmatrix}7.82\\-3.63\\4.01\end{bmatrix} = 7.82\begin{bmatrix}1\\-0.46\\0.51\end{bmatrix} = \lambda_3 X_3$$ $$AX_3 = \begin{bmatrix}7.94\\-3.89\\3.99\end{bmatrix} = 7.94\begin{bmatrix}1\\-0.49\\0.5\end{bmatrix} = \lambda_4 X_4$$ $$AX_4 = \begin{bmatrix}7.98\\-3.97\\3.99\end{bmatrix} = 7.98\begin{bmatrix}1\\-0.5\\0.5\end{bmatrix} = \lambda_5 X_5$$ $$AX_5 = \begin{bmatrix}8\\-4\\4\end{bmatrix} = 8\begin{bmatrix}1\\-0.5\\0.5\end{bmatrix} = \lambda_6 X_6$$ Thus the dominant eigenvalue is 8 and corresponding eigenvector is $\begin{bmatrix}1\\-0.5\\0.5\end{bmatrix}$	1 1 1 1 1
3a)	For the curve $r = a\log\theta$ $$\frac{dr}{d\theta} = \frac{a}{\theta}$$ $$\tan\phi_1 = r\frac{d\theta}{dr} = \theta\log\theta$$	2
	For the curve $r = \frac{a}{\log\theta}$ $$\frac{dr}{d\theta} = -\frac{a}{(\log\theta)^2}\cdot\frac{1}{\theta}$$	2
	$$\tan\phi_2 = -\theta\log\theta$$ Angle of intersection: $\tan\lvert\phi_1 - \phi_2\rvert = \left\lvert\frac{\tan\phi_1 - \tan\phi_2}{1+\tan\phi_1.\tan\phi_2}\right\rvert = \left\lvert\frac{2\theta\log\theta}{1-\theta^2(\log\theta)^2}\right\rvert --(1)$	1
	To find θ: At the point of intersection, $a\log\theta = \frac{a}{\log\theta} \Rightarrow (\log\theta)^2 = 1$ $$\therefore \theta = e$$	2
	$$(1) \Rightarrow \tan\lvert\phi_1 - \phi_2\rvert = \frac{2e}{1-e^2}$$ $$\therefore \lvert\phi_1 - \phi_2\rvert = \tan^{-1}\left(\frac{2e}{1-e^2}\right) = 2\tan^{-1}e$$	1
3b)	$$\frac{dx}{dt} = a\left(-\sin t + \frac{1}{\tan\frac{t}{2}}.\sec^2\frac{t}{2}\cdot\frac{1}{2}\right) = \frac{a\cos^2 t}{\sin t} = a\cos t\cot t$$	2
	$$\frac{dy}{dt} = a\cos t$$	1
	$$y_1 = \frac{dy}{dx} = \frac{\frac{dy}{dt}}{\frac{dx}{dt}} = \tan t$$	1

	$y_2 = \frac{d^2y}{dx^2} = \frac{d}{dx}\left(\frac{dy}{dx}\right) = \frac{d}{dt}\left(\frac{dy}{dx}\right)\left(\frac{dt}{dx}\right) = \frac{1}{a}\sec^4 t \sin t$ $\rho = \frac{(1+y_1^2)^{\frac{3}{2}}}{y_2} = \frac{(1+\tan^2 t)^{\frac{3}{2}}}{\frac{1}{a}\sec^4 t \sin t} = \frac{a\sec^3 t}{\sec^4 t \sin t} = a\cot t$	2 2
4a)	$r = a(1+\sin\theta)$ Apply log on both sides $\log r = \log a + \log(1+\sin\theta)$ Differentiating both sides w.r.t θ $\frac{1}{r}\frac{dr}{d\theta} = \frac{\cos\theta}{1+\sin\theta}$ $\cot\phi = \frac{\cos^2\frac{\theta}{2} - \sin^2\frac{\theta}{2}}{\cos^2\frac{\theta}{2} + \sin^2\frac{\theta}{2} + 2\sin\frac{\theta}{2}\cos\frac{\theta}{2}} = \frac{1-\tan\frac{\theta}{2}}{1+\tan\frac{\theta}{2}}$ $\cot\phi = \frac{\tan\frac{\pi}{4} - \tan\frac{\theta}{2}}{\tan\frac{\pi}{4} + \tan\frac{\theta}{2}} = \tan\left(\frac{\pi}{4} - \frac{\theta}{2}\right) = \cot\left(\frac{\pi}{2} - \frac{\pi}{4} + \frac{\theta}{2}\right)$ $\therefore \phi = \frac{\pi}{4} + \frac{\theta}{2}$	1 2 2 2 1
4b)	$y = \log(1+e^x) \Rightarrow y(0) = \log 2$ $y_1 = \frac{e^x}{1+e^x} \Rightarrow y_1(0) = \frac{1}{2}$ $y_2 = \frac{e^x}{(1+e^x)^2} = \frac{e^x}{1+e^x}\cdot\frac{1}{1+e^x} = y_1(1-y_1)$ $y_2(0) = \frac{1}{2}\left(1-\frac{1}{2}\right) = \frac{1}{4}$ $y_3 = y_1(-y_2) + (1-y_1)y_2 = y_2 - 2y_1y_2 \Rightarrow y_3(0) = \frac{1}{4} - 2.\frac{1}{2}.\frac{1}{4} = 0$ $y_4 = y_3 - 2(y_1y_3 + y_2^2) \Rightarrow y_4(0) = -\frac{1}{8}$ $\therefore \log(1+e^x) = \log 2 + \frac{x}{2} + \frac{x^2}{8} - \frac{x^4}{192}$	1 2 2 2 1
5a)	Differentiating θ partially wrt r $\frac{\partial\theta}{\partial r} = t^n e^{-\frac{r^2}{4t}}\left(-\frac{2r}{4t}\right) = -\frac{r}{2}t^{n-1}e^{-\frac{r^2}{4t}}$ $\therefore r^2\frac{\partial\theta}{\partial r} = -\frac{r^3}{2}t^{n-1}e^{-\frac{r^2}{4t}}$ $\frac{\partial}{\partial r}\left(r^2\frac{\partial\theta}{\partial r}\right) = -\frac{3}{2}r^2t^{n-1}e^{-\frac{r^2}{4t}} + \left(-\frac{r^3}{2}\right)t^{n-1}e^{-\frac{r^2}{4t}}\left(-\frac{2r}{4t}\right)$ $\frac{1}{r^2}\frac{\partial}{\partial r}\left(r^2\frac{\partial\theta}{\partial r}\right) = \left(-\frac{3}{2}t^{n-1} + \frac{r^2}{4}t^{n-2}\right)e^{-\frac{r^2}{4t}} - - - (1)$ Differentiating θ partially wrt t $\frac{\partial\theta}{\partial t} = \left(nt^{n-1} + \frac{1}{4}r^2t^{n-2}\right)e^{-\frac{r^2}{4t}} - - - (2)$ Given: $\frac{1}{r^2}\frac{\partial}{\partial r}\left(r^2\frac{\partial\theta}{\partial r}\right) = \frac{\partial\theta}{\partial t}$ $\left(-\frac{3}{2}t^{n-1} + \frac{r^2}{4}t^{n-2}\right)e^{-\frac{r^2}{4t}} = \left(nt^{n-1} + \frac{1}{4}r^2t^{n-2}\right)e^{-\frac{r^2}{4t}}$	1 1 2 2 1

7a)		2
	$\int_0^1 \int_{\sqrt{x}}^1 \sqrt{1+y^3}\ dy\ dx = \int_{y=0}^1 \int_{x=0}^{y^2} \sqrt{1+y^3}\ dx\ dy$	2
	$= \int_{y=0}^1 \left\{x\sqrt{1+y^3}\right\}_{x=0}^{y^2} dy = \int_0^1 y^2\sqrt{1+y^3}\ dy$	1
	Let $1+y^3 = t \Rightarrow 3y^2dy = dt \Rightarrow y^2dy = \frac{dt}{3}$ When $y = 0 \Rightarrow t = 1, y = 1 \Rightarrow t = 2$	1
	$\int_0^1 \int_{\sqrt{x}}^1 \sqrt{1+y^3}\ dy\ dx = \frac{1}{3}\int_{t=1}^2 \sqrt{t}\ dt = \frac{2}{9}\left(2^{\frac{3}{2}} - 1\right) = \frac{2}{9}\left(2\sqrt{2} - 1\right)$	2
7b)		2
	$\int_{-1}^1 \int_{-\sqrt{1-x^2}}^0 \cos(x^2+y^2)\ dy\ dx = \int_{\theta=\pi}^{2\pi} \int_{r=0}^1 \cos(r^2)\, r\ dr\ d\theta$	2
	Let $r^2 = t \Rightarrow rdr = \frac{dt}{2}$	1
	When $r = 0 \Rightarrow t = 0, r = 1 \Rightarrow t = 1$ $\int_{-1}^1 \int_{-\sqrt{1-x^2}}^0 \cos(x^2+y^2)\ dy\ dx = \frac{1}{2}\int_{\theta=\pi}^{2\pi} \int_0^1 \cos(t)\ dt\ d\theta$	2
	$= \frac{1}{2}\int_{\pi}^{2\pi} \{\sin t\}_0^1\ d\theta = \frac{1}{2}\sin 1\ \{\theta\}_{\pi}^{2\pi} = \frac{\pi}{2}\sin 1$	1
8a)		2

	$\left(n+\frac{3}{2}\right)t^{n-1}e^{-\frac{r^2}{4t}}=0 \Rightarrow n=-\frac{3}{2}$	
5b)	The distance from a point (x,y,z) to the point $(3,1,-1)$ is $d=\sqrt{(x-3)^2+(y-1)^2+(z+1)^2}$ $d^2=f(x,y,z)=(x-3)^2+(y-1)^2+(z+1)^2$ The constraint is that the point (x,y,z) lies on sphere, that is, $g(x,y,z)=x^2+y^2+z^2=4$ According to the method of Lagrange multipliers, $2(x-3)=2x\lambda$ $2(y-1)=2y\lambda$ $2(z+1)=2z\lambda$ Solve x,y,z in terms of λ $x=\frac{3}{1-\lambda}, y=\frac{1}{1-\lambda}, z=-\frac{1}{1-\lambda}---(1)$ Given: $x^2+y^2+z^2=4 \Rightarrow \left(\frac{3}{1-\lambda}\right)^2+\left(\frac{1}{1-\lambda}\right)^2+\left(\frac{1}{1-\lambda}\right)^2=4$ $\Rightarrow (1-\lambda)^2=\frac{11}{4} \Rightarrow \lambda=1\pm\frac{\sqrt{11}}{2}$ $(1) \Rightarrow x=\frac{6}{\sqrt{11}}, y=\frac{2}{\sqrt{11}}, z=-\frac{2}{\sqrt{11}}$ and $x=-\frac{6}{\sqrt{11}}, y=-\frac{2}{\sqrt{11}}, z=\frac{2}{\sqrt{11}}$ Closest point: $\left(\frac{6}{\sqrt{11}},\frac{2}{\sqrt{11}},-\frac{2}{\sqrt{11}}\right)$ and Farthest point : $\left(-\frac{6}{\sqrt{11}},-\frac{2}{\sqrt{11}},\frac{2}{\sqrt{11}}\right)$	2 1 1 2 2
6a)	$\frac{\partial y}{\partial t}=a\,f'(x+at)-a\,g'(x-at)$ $\frac{\partial^2 y}{\partial t^2}=a^2 f''(x+at)+a^2 g''(x-at)=a^2\left(f''(x+at)+g''(x-at)\right)---(1)$ $\frac{\partial y}{\partial x}=f'(x+at)+g'(x-at)$ $\frac{\partial^2 y}{\partial x^2}=f''(x+at)+g''(x-at)---(2)$ From (1) and (2) $\frac{\partial^2 y}{\partial t^2}=a^2\frac{\partial^2 y}{\partial x^2}$	2 2 1 2 1
6b)	Let x, y, z be the edges of the box and S be its surface. Then $S=xy+2yz+2zx$ and $xyz=32$ By Lagrange's method of multiplier $F(x,y,z)=(xy+2yz+2zx)+\lambda(xyz-32)$ $F_x=0 \Rightarrow y+2z+\lambda(yz)=0,$ $F_y=0 \Rightarrow x+2z+\lambda(xz)=0,$ $F_z=0 \Rightarrow 2y+2z+\lambda(xy)=0$ $\frac{y+2z}{yz}=\frac{(x+2z)}{xz}=\frac{(2y+2x)}{xy}$ $\Rightarrow x=y=2z$ Given: $xyz=32 \Rightarrow \frac{x^2x}{2}=32 \Rightarrow x=4$ $\Rightarrow y=4, z=2$	1 3 2 2

	$y_p = e^{3x}\left(\frac{x^4}{12}+\frac{7x^3}{6}+\frac{5x^2}{2}\right)$ $y_p = \frac{e^{3x}}{12}(x^4+14x^3+30x^2)$ The general solution is $y = y_c + y_p = (c_1+c_2x)e^{3x} + \frac{e^{3x}}{12}(x^4+14x^3+30x^2)$	1 1
9b)	The auxiliary equation is $m^2+16=0$ $\Rightarrow m = \pm 4i$ Complementary function: $y_c = c_1\cos 4x + c_2\sin 4x --(1)$ To find particular integral: Let $y_p = A(x)\cos 4x + B(x)\sin 4x --(2)$ $W(y_1,y_2) = \begin{vmatrix}\cos 4x & \sin 4x\\ -4\sin 4x & 4\cos 4x\end{vmatrix} = 4$ $A(x) = \int -\frac{g(x)y_2(x)}{a_0.W(y_1,y_2)}dx = -\int \frac{32\sec 2x\,.\sin 4x}{1.4}dx$ $A(x) = -8\int \frac{1}{\cos 2x}2\sin 2x\cos 2x\ dx = -16\frac{(-\cos 2x)}{2} = 8\cos 2x$ $B(x) = \int \frac{g(x)y_1(x)}{a_0.W(y_1,y_2)}dx = \int \frac{32\sec 2x\,.\cos 4x}{1.4}dx$ $B(x) = 8\int \frac{\cos 4x}{\cos 2x}dx = 8\int \frac{(2\cos^2 2x-1)}{\cos 2x}dx$ $B(x) = 8\int(2\cos 2x - \sec 2x)\,dx = 8\left[\sin 2x - \frac{1}{2}\log(\sec 2x+\tan 2x)\right]$ $(2)\Rightarrow y_p = 8\cos 2x\cos 4x + 8\left[\sin 2x - \frac{1}{2}\log(\sec 2x+\tan 2x)\right]\sin 4x$ $y_p = 8\cos 2x - 4\sin 4x\ \log(\sec 2x+\tan 2x)$ The general solution is : $y = y_c + y_p = c_1\cos 4x + c_2\sin 4x + 8\cos 2x - 4\sin 4x\ \log(\sec 2x+\tan 2x)$	2 1 2 2 1
10a)	The given equation is 2nd order Cauchy's equation $Let\ x = e^z \Rightarrow z = \log x$ $x^2\frac{d^2y}{dx^2} = D_1(D_1-1)y, x\frac{dy}{dx} = D_1y$ where $D_1 = \frac{d}{dz}$ The given equation reduces to $[D_1(D_1-1)-D_1+2]y = e^z z$ $(D_1^2-2D_1+2)y = e^z z$ The auxiliary equation is $m^2-2m+2=0$ $m = \frac{2\pm\sqrt{4-8}}{2} = 1\pm i$ $y_c = x(c_1\cos z + c_2\sin z) = x[c_1\cos(\log x)+c_2\sin(\log x)]$ $y_p = \frac{1}{D_1^2-2D_1+2}z.e^z = e^z\frac{1}{(D_1+1)^2-2(D_1+1)+2}z$ $y_p = e^z.\frac{1}{D_1^2+1}z = e^z(1+D_1^2)^{-1}z = e^z(1-D_1^2+\cdots)z = ze^z = x\log x$ The general solution is $y = y_c + y_p = x[c_1\cos(\log x)+c_2\sin(\log x)] + x\log x$	1 1 2 4
10b)	Given: $\frac{d^2x}{dt^2}+4\frac{dx}{dt}+20x = 10\cos 2t$ The auxiliary equation is $m^2+4m+20=0$	

	$$Area = \int_{\frac{\pi}{2}}^{\pi} \int_{a\sin\theta}^{a(1+\cos\theta)} r\,dr\,d\theta$$	1
	$$A = \int_{\frac{\pi}{2}}^{\pi} \frac{a^2}{2}(\sin^2\theta + 1 + 2\cos\theta + \cos^2\theta)d\theta$$	1
	$$A = \int_{\frac{\pi}{2}}^{\pi} \frac{a^2}{2}(1 + 2\cos\theta + \cos 2\theta)d\theta$$	1
	$$A = \frac{a^2}{2}\left[\theta + 2\sin\theta + \frac{\sin 2\theta}{2}\right]_{\frac{\pi}{2}}^{\pi} = \frac{a^2}{2}\left[(\pi + 0 + 0) - \left(\frac{\pi}{2} + 2 + 0\right)\right]$$	2
	$$\therefore Area = \frac{a^2}{2}\left(2 + \frac{\pi}{2}\right) = a^2\left(1 + \frac{\pi}{4}\right)\ sq.units$$	1
8b)	Mass$= \iiint \mu\, xyz\, dx\, dy\, dz$	
	$$Mass = \int_{x=0}^{a} \int_{y=0}^{b\left(1-\frac{x}{a}\right)} \int_{z=0}^{c\left(1-\frac{x}{a}-\frac{y}{b}\right)} \mu\, xyz\ dz\, dy\, dx$$	1
	$$Mass = \mu \int_{0}^{a} \int_{0}^{b\left(1-\frac{x}{a}\right)} xy\left[\frac{z^2}{2}\right]_0^{c\left(1-\frac{x}{a}-\frac{y}{b}\right)} dy\, dx$$	1
	$$Mass = \mu \int_{0}^{a} \int_{0}^{b\left(1-\frac{x}{a}\right)} xy\frac{c^2}{2}\left(1-\frac{x}{a}-\frac{y}{b}\right)^2 dy\, dx$$	1
	$$Mass = \frac{\mu c^2}{2} \int_{0}^{a} \int_{0}^{b\left(1-\frac{x}{a}\right)} x.\left[\left(1-\frac{x}{a}\right)^2 y - 2\left(1-\frac{x}{a}\right)\frac{y^2}{b} + \frac{y^3}{b^2}\right] dy\, dx$$	1
	$$Mass = \frac{\mu c^2}{2} \int_{0}^{a} x\left[\left(1-\frac{x}{a}\right)^2 \frac{y^2}{2} - 2\left(1-\frac{x}{a}\right)\frac{y^3}{3b} + \frac{y^4}{4b^2}\right]_0^{b\left(1-\frac{x}{a}\right)} dx$$	1
	$$Mass = \frac{\mu c^2}{2} \int_{0}^{a} x\left[\frac{b^2}{2}\left(1-\frac{x}{a}\right)^4 - \frac{2b^2}{3}\left(1-\frac{x}{a}\right)^4 + \frac{b^2}{4}\left(1-\frac{x}{a}\right)^4\right] dx$$	1
	$$Mass = \frac{\mu b^2 c^2}{24} \int_{0}^{a} x\left(1-\frac{x}{a}\right)^4 dx = \frac{\mu a^2 b^2 c^2}{720}.$$	2
9a)	The auxiliary equation is $m^2 - 6m + 9 = 0$	1
	$\Rightarrow m = 3, 3$	1
	Complementary function: $y_c = (c_1 + c_2 x)e^{3x}$	1
	Particular integral: $y_p = \frac{1}{D^2-6D+9} e^{3x}(x^2 + 7x + 5)$	
	$$y_p = e^{3x} \frac{1}{(D+3)^2 - 6(D+3) + 9}(x^2 + 7x + 5)$$	1
	$$y_p = e^{3x} \frac{1}{D^2}(x^2 + 7x + 5) = e^{3x} \frac{1}{D}\frac{1}{D}(x^2 + 7x + 5)$$	1
	$$y_p = e^{3x} \frac{1}{D}\left(\frac{x^3}{3} + \frac{7x^2}{2} + 5x\right)$$	1

	$\Rightarrow m = -2 \pm 4i$	
	$x_c = e^{-2t}(c_1 \cos 4t + c_2 \sin 4t)$	2
	$x_p = \frac{1}{D^2 + 4D + 20} 10 \cos 2t = \frac{10}{-4 + 4D + 20} \cos 2t$	
	$x_p = \frac{10}{16 + 4D} \cos 2t = 10 \frac{16 - 4D}{256 + 16D^2} \cos 2t = \frac{10}{320}(16 - 4D) \cos 2t$	
	$x_p = \frac{1}{2} \cos 2t + \frac{1}{4} \sin 2t$	2
	$\therefore x = x_c + x_p = e^{-2t}(c_1 \cos 4t + c_2 \sin 4t) + \frac{1}{2} \cos 2t + \frac{1}{4} \sin 2t - -(1)$	
	$x' = e^{-2t}(-4c_1 \sin 4t + 4c_2 \cos 4t) - 2e^{-2t}(c_1 \cos 4t + c_2 \sin 4t) - \sin 2t$	
	$+ \frac{1}{2} \cos 2t - -(2)$	1
	Applying initial conditions: $x(0) = 0, x'(0) = 0$	
	$x(0) = 0 \Rightarrow 0 = c_1 + \frac{1}{2} \Rightarrow c_1 = -\frac{1}{2}$	1
	$x'(0) = 0 \Rightarrow 0 = 4c_2 - 2c_1 + \frac{1}{2} \Rightarrow c_2 = -\frac{3}{8}$	1
	The particular solution is	
	$x = e^{-2t}\left(-\frac{1}{2} \cos 4t - \frac{3}{8} \sin 4t\right) + \frac{1}{2} \cos 2t + \frac{1}{4} \sin 2t$	1

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RV COLLEGE OF ENGINEERING®

(An Autonomous Institution Affiliated to VTU)

I Semester B. E. Supplementary Examinations Oct-2023

(Common to AS, CH, IM, ME)

FUNDAMENTALS OF LINEAR ALGEBRA, CALCULUS AND DIFFERENTIAL EQUATIONS

Time: 03 Hours *Maximum Marks: 100*

Instructions to candidates:

1. Answer all questions from Part A. Part A questions should be answered in first three pages of the answer book only.
2. Answer FIVE full questions from Part B. In Part B question number 2 is compulsory. Answer any one full question from 3 and 4, 5 and 6, 7 and 8, 9 and 10.
3. Use of scientific calculator is permitted.
4. Use of mathematics Handbook is permitted. Do not write anything on handbook.

PART-A

1	1.1	If two eigen values of $\begin{bmatrix} 8 & -6 & 2 \\ -6 & 7 & -4 \\ 2 & -4 & 3 \end{bmatrix}$ are 3 and 15, then the third eigen value is _____.	01
	1.2	The maximum value of the rank of 4×5 matrix is _____.	01
	1.3	If $1, 1/2$ are the eigen values of A, the eigen values of A^T= _____ and eigen values of A^{-1} = _____.	02
	1.4	The radius of curvature of the curve $y = e^x$ at the point where it crosses the y-axis is _____.	02
	1.5	Maclaurin's series expansion of $y = a^x$ is _____.	02
	1.6	If $u = x^y$, then $\frac{\partial^2 u}{\partial x^2}$ = _____	02
	1.7	If $x = r\cos\theta$, $y = r\sin\theta$ then $\frac{\partial(x,y)}{\partial(r,\theta)}$ is equal to _____.	02
	1.8	Change to polar coordinates $\int_{-a}^{a}\int_{0}^{\sqrt{a^2-x^2}} f(x,y)dx\,dy$ =_____.	02
	1.9	The solution of $(D^2 - 10D + 25)y = 0$ is _____.	02
	1.10	The Particular integral of $(D^3 + 1)y = e^{3x}$ is _____.	02
	1.11	Evaluate $\int_0^1\int_0^2 x^2y\,dy\,dx$	02

PART-B

2	a	Find the values of a and b for which the system $2x + 3y + 5z = 9$, $7x + 3y - 2z = 8$, $2x + 3y + az = b$ has: i) A unique solution ii) Infinitely many solutions iii) No solution.	05
	b	Apply Gauss-Seidel iteration to solve the system of equations $12x_1 + 3x_2 - 5x_3 = 1$; $x_1 + 5x_2 + 3x_3 = 28$; $3x_1 + 7x_2 + 13x_3 = 76$. Carry out four iterations, starting with the initial approximations $(1, 0, 1)$.	05
	c	Find the largest eigen values and its corresponding eigen vector of the matrix $A = \begin{bmatrix} -2 & 0 & -1 \\ 1 & -1 & 1 \\ 2 & 2 & 0 \end{bmatrix}$ with initial eigen vector $[1 \quad 1 \quad 1]^T$. Perform 5 iterations.	06

3	a	Find the angle of intersection of the curves $r = a\log_e\theta$, $r = a/\log_e\theta$.	
	b	Obtain the radius of curvature of the curve $x = a(\theta + sin\theta)$, $y = a(1 - \cos\theta)$.	
		OR	
4	a	Find the centre of curvature at the point $(1,4)$ of the curve $y = 2x^3 - x + 3$.	08
	b	Obtain Maclaurin's series expansion of $\tan^{-1}x$, hence prove that $\sin^{-1}\left(\frac{2x}{1+x^2}\right) = 2\{x - \frac{x^3}{3} + \cdots\}$	08
5	a	The temperature T at any point (x,y,z) in space is $400xy^2z$. Find the highest temperature at the surface of unit sphere $x^2 + y^2 + z^2 = 1$ using method of Lagrange multiplier.	08
	b	Evaluate $J = \frac{\partial(x,y,z)}{\partial(u,v,w)}$ for the function $u = x + y + z$, $uv = y + z$, $uvw = z$.	08
		OR	
6	a	If $u = f\left(\frac{x}{y}, \frac{y}{z}, \frac{z}{x}\right)$ show that $x\frac{\partial u}{\partial x} + y\frac{\partial u}{\partial y} + z\frac{\partial u}{\partial z} = 0$.	08
	b	If $z = \frac{x^2+y^2}{x+y}$ prove that $\left(\frac{\partial z}{\partial x} - \frac{\partial z}{\partial y}\right)^2 = 4\left(1 - \frac{\partial z}{\partial x} - \frac{\partial z}{\partial y}\right)$.	08
7	a	Change the order of integration $I = \int_0^1 \int_{x^2}^{2-x} xy\, dy\, dx$ and hence evaluate the integral.	08
	b	Find by triple integration, the volume of the sphere $x^2 + y^2 + z^2 = a^2$	08
		OR	
8	a	Evaluate the integral by changing into polar coordinates: $\int_0^2 \int_0^{\sqrt{4-x^2}} (x^2y + y^3)\, dydx$	08
	b	Find the area between the parabolas $y^2 = 2x$ and $x^2 = 2y$ using double integration.	08
9	a	Solve $(D^2 - 4D + 4)y = \cos^2 x + 3^x$.	08
	b	Solve $x^3y''' + 3x^2y'' + 2xy' = x + \log_e x$.	08
		OR	
10	a	Solve $y'' - 2y' + y = e^x \log_e x$ using method of variation of parameters.	08
	b	The charge $q = q(t)$ on the capacitor in a LRC series circuit is governed by the equation: $L\frac{d^2q}{dt^2} + R\frac{dq}{dt} + \frac{1}{c}q = E(t)$. Find the charge on the capacitor $q = q(t)$ when $L = 0.25h$, $R = 10\Omega$, $C = 0.001f$, $E(t) = 0$, $q(0) = q_0$, $\frac{dq}{dt} = 0$ for $t = 0$.	08

COURSE CODE:
Question No

SCHEME AND SOLUTION April-2023

UG

COURSE CODE: 22mAI13 **COURSE:** FUNDAMENTALS OF LINEAR ALGEBRA, CALCULUS AND DIFF. EQUATIONS

Question No			Marks
1.	1.1	0	2
	1.2	4	2
	1.3	~~1/2~~ 1, 1/2 & 1, 2	2
	1.4	$2\sqrt{2}$	2
	1.5	~~$\bar{x} = x - \frac{y_1(1+y_1^2)}{y_2}$~~ ~~$\bar{y} = y + \frac{1+y_1^2}{y_2}$~~	2
	1.6	yx^{y-1} , $y(y-1)x^{y-2}$.	2
	1.7	r	2
	1.8	$\int_{\theta=-\pi}^{\pi}\int_{r=0}^{a} f(r,\theta)\, r\, dr\, d\theta$	2
	1.9	$y = (C_1 + C_2 x)e^{5x}$	2
	Fig 10	$y_p = \frac{e^{3x}}{28}$	2
	1.11	$\int_0^1\int_0^2 x^2 y\, dy\, dx = \int_0^1 \frac{x^3}{3}\Big|_0^1 \quad \frac{y^2}{2}\Big|_0^2 = \frac{1}{3}(2) = \frac{2}{3}$	
	1.5	$y = a^x$, $y' = a^x \log a$, $y'' = a^x(\log a)^2$, $y''' = a^x(\log a)^3$	
		$y = f(0) + \frac{x}{1!}f'(0) + \frac{x^2}{2!}f''(0) + \frac{x^3}{3!}f'''(0) + \cdots$	
		$y = 1 + \log a\, x + \frac{x^2}{2}(\log a)^2 + \frac{x^3}{3!}(\log a)^3 + \cdots$	

SCHEME AND SOLUTION **April-2023**

COURSE CODE: 22MAI1B **COURSE:** FUNDAMENTALS OF LA, CALCULUS & D.E

UG

Question No	PART-B	Marks
2 a)	$[A:B] = \begin{bmatrix} 2 & 3 & 5 & : & 9 \\ 7 & 3 & -2 & : & 8 \\ 2 & 3 & a & : & b \end{bmatrix} \sim \begin{bmatrix} 2 & 3 & 5 & : & 9 \\ 0 & -15 & -39 & : & -47 \\ 0 & 0 & a-5 & : & b-9 \end{bmatrix}$	2M
	(i) $a \neq 5$, unique solution	1M
	(ii) $a=5$, $b=9$, infinitely many solutions	1M
	(iii) $a=5$, $b \neq 9$, no solution.	1M
b)	$x_1 = \frac{1}{12}[1-3x_2+5x_3]$ $x_2 = \frac{1}{5}[28-x_1-3x_3]$ $x_3 = \frac{1}{13}[76-3x_1-7x_2]$	1M
	Given $x_1^0 = 1$, $x_2^0 = 0$, $x_3^0 = 1$	
	First iteration: $x_1 = 0.5$, $x_2 = 4.9$, $x_3 = 3.0923$	1M
	Second iteration: $x_1 = 0.1467$, $x_2 = 3.7153$, $x_3 = 3.8118$	1M
	Third iteration: $x_1 = 0.0901$, $x_2 = 3.7205$, $x_3 = 3.915$	1M
	$\therefore x_1 = 1$, $x_2 = 3$, $x_3 = 4$	1M
c)	$Ax^0 = \begin{bmatrix} -2 & 0 & -1 \\ 1 & -1 & 1 \\ 2 & 2 & 0 \end{bmatrix} \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix} = 4 \begin{bmatrix} -0.75 \\ 0.25 \\ 1 \end{bmatrix}$	1M
	$Ax^1 = \begin{bmatrix} 0.5 \\ 0 \\ -1 \end{bmatrix} = 1 \begin{bmatrix} 0.5 \\ 0 \\ -1 \end{bmatrix}$	1M
	$Ax^2 = 1 \begin{bmatrix} 0 \\ -0.5 \\ 1 \end{bmatrix}$	1M
	$Ax^3 = 1.5 \begin{bmatrix} -0.67 \\ 1 \\ -0.67 \end{bmatrix}$ $\therefore \lambda = 2.34$	1M
	$Ax^4 = 2.34 \begin{bmatrix} 0.86 \\ -1 \\ 0.28 \end{bmatrix}$ $X = \begin{bmatrix} 0.86 \\ -1 \\ 0.28 \end{bmatrix}$ —1M	1M

COURSE CODE: **COURSE:**

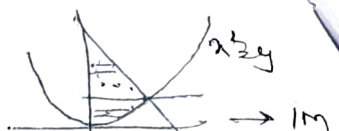
Question No		
3	a) $r = a\log\theta$, $r = \frac{a}{\log\theta}$ Equating $\theta = e$ is the point of intersection	1m
	At $\theta = \theta_1$, $\tan\phi_1 = e$, $\tan\phi_2 = -e$	2m+2m
	Now $\tan\phi = \tan\lvert\phi_1 - \phi_2\rvert = \left\lvert \frac{\tan\phi_1 - \tan\phi_2}{1+\tan\phi_1\tan\phi_2} \right\rvert = \left\lvert \frac{2e}{1-e^2} \right\rvert$	3m
	b) $y_1 = \frac{dy/d\theta}{dx/d\theta} \frac{a\sin\theta}{a(1+\cos\theta)} = \tan\theta/2$	2m
	$y_2 = \frac{1}{4a}\sec^4(\theta/2)$	3m
	$\rho = \frac{4a}{\sec(\theta/2)} = 4a\cos\theta/2$	3m
4	a) $y = 2x^3 - x + 3$	
	$y_1(1,4) = 5$	1m
	$y_2(1,4) = 12$	1m
	$\bar{x}_{(1,4)} = -9.83$ $\bar{y}_{(1,4)} = 6.166$ $\rho_{(1,4)} = 11.047$	1m+1m+2m
	$(a+9.83)^2 + (y-6.166)^2 = (11.047)^2$	2m
	b) Put $x = \tan\theta$ we get $\sin^{-1}\left(\frac{2x}{1+x^2}\right) = 2\theta$, $\theta = \tan^{-1}x$	2M
	$y = \tan^{-1}x$ $y(0) = 0$	1M
	$y_1 = \frac{1}{1+x^2}$ $y_1(0) = 1$	1m
	$y_2 = -\frac{2x}{(1+x^2)^2}$ $y_2(0) = 0$, $y_3(0) = 2$	1m 1m
	By Maclaurin's series we get $\tan^{-1}(x) = x - \frac{x^3}{3} + \ldots$	1m
	$\therefore \sin^{-1}\left(\frac{2x}{1+x^2}\right) = 2\left\{x - \frac{x^3}{3} + \ldots\right\}$	1m

SCHEME AND SOLUTION April-2023

COURSE CODE: 22MAILD **COURSE:** Fundamentals of LA, Calculus & D.E UG

Question No		Marks
5	a) $F(x,y,z) = f(x,y,z) + \lambda\phi(x,y,z) = 400xy^2z + \lambda(x^2+y^2+z^2-1)$	2m
	By Lagrange multipliers, $\frac{\partial F}{\partial x}=0$, $\frac{\partial F}{\partial y}=0$, $\frac{\partial F}{\partial z}=0$	
	$400y^2z + 2\lambda x = 0$	2m
	$800xyz + \lambda 2y = 0$	
	$400xy^2 + 2\lambda z = 0$ Solving we get $y^2 = 2x^2$, $2z^2 = y^2$	2m
	Sub in $x^2+y^2+z^2=1 \Rightarrow x = \pm 1/2 \Rightarrow z = \pm\frac{1}{2}$, $y = \pm\frac{1}{\sqrt{2}}$	2m
	$\therefore T = 400xy^2z = 400\left(\frac{1}{2}\right)\left(\frac{1}{2}\right)\left(\frac{1}{2}\right) = 50$	
	b) $u = x+y+z$, $uv = y+z$, $uvw = z$	
	By Substitution $x = u-uv$, $y = uv(1-w)$, $z = uvw$	3m
	$J = \begin{vmatrix} 1-v & -u & 0 \\ v-vw & u-uw & -uv \\ vw & uw & uv \end{vmatrix} = u^2v$	3m 2m
6	a) Let $r = \frac{x}{y}$, $s = \frac{y}{z}$, $t = \frac{z}{x}$	1m
	$\frac{\partial u}{\partial x} = \frac{\partial u}{\partial r}\frac{1}{y} + \frac{\partial u}{\partial t}\left(\frac{-z}{x^2}\right)$	2m
	$\frac{\partial u}{\partial y} = \frac{\partial u}{\partial r}\left(\frac{-x}{y^2}\right) + \frac{\partial u}{\partial s}\left(\frac{1}{z}\right)$	2m
	$\frac{\partial u}{\partial z} = \frac{\partial u}{\partial s}\left(\frac{-y}{z^2}\right) + \frac{\partial u}{\partial t}\left(\frac{1}{x}\right)$ $\therefore xu_x + yu_y + zu_z = 0$	2m 1m
	b) $z = \frac{x^2+y^2}{x+y}$	
	$\frac{\partial z}{\partial x} = \frac{x^2+2xy-y^2}{(x+y)^2}$	2m
	$\frac{\partial z}{\partial y} = \frac{y^2+2xy-x^2}{(x+y)^2}$	2m
	$\left(\frac{\partial z}{\partial x} - \frac{\partial z}{\partial y}\right)^2 = 4\left(\frac{x-y}{x+y}\right)^2$ $4\left(1 - \frac{\partial z}{\partial x} - \frac{\partial z}{\partial y}\right) = 4\left(\frac{x-y}{x+y}\right)^2$	2m
	$\therefore \left(\frac{\partial z}{\partial x} - \frac{\partial z}{\partial y}\right)^2 = 4\left(1 - \frac{\partial z}{\partial x} - \frac{\partial z}{\partial y}\right)$	2m

7 a) $I = \int_0^1 \int_0^{\sqrt{y}} xy\,dx\,dy + \int_1^2 \int_0^{2-y} xy\,dx\,dy$ → 2m



$= \int_0^1 y\left(\frac{x^2}{2}\right)_0^{\sqrt{y}} dy + \int_1^2 y\left[\frac{x^2}{2}\right]_0^{2-y} dy$ → 2m

$= \frac{1}{2}\int_0^1 y^2 dy + \frac{1}{2}\int_1^2 y(2-y^2)\,dy = \frac{3}{8}$ → 3m

b) $V = 8\int_0^a \int_0^{\sqrt{a^2-y^2}} \int_0^{\sqrt{a^2-x^2-y^2}} dz\,dy\,dx$ → 2m

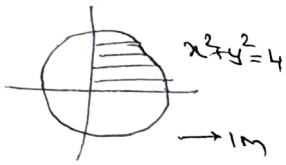
Put $x = r\cos\theta$, $y = r\sin\theta$ ∴ $dx\,dy = r\,dr\,d\theta$ → 1m

$V = 8\int_0^{\pi/2} \int_0^a \sqrt{a^2-r^2}\, r\,dr\,d\theta$ — 2m

$= \frac{8}{2}\int_0^{\pi/2} \int_0^{a^2} \sqrt{t}\,dt\,d\theta$ Put $a^2-r^2=t$ — 2m

$= \frac{4\pi}{3}a^3$ → 1m

8 a) $I = \int_0^{\pi/2} \int_0^2 (r^2\cos^2\theta\sin\theta \cdot r + r^3\sin^3\theta)\, r\,dr\,d\theta$



$= \int_0^{\pi/2} \int_0^2 (\cos^2\theta\sin\theta + \sin^3\theta)\, r^4\,dr\,d\theta$

$= \int_0^{\pi/2} (\cos^2\theta\sin\theta + \sin^3\theta)\left[\frac{r^5}{5}\right]_0^2 d\theta = \frac{32}{5}$

b) $I = \int_{-\sqrt{2}}^{\sqrt{2}} \int_{y^2}^{4-y^2} dx\,dy = \int_{-\sqrt{2}}^{\sqrt{2}} [4-y^2-y^2]\,dy = \int_{-\sqrt{2}}^{\sqrt{2}} [4-2y^2]\,dy = \frac{16\sqrt{2}}{3}$

→ 2m → 2m → 2m → 2m

9 a) $m^2-4m+4=0$, $m=2,2$. C.F $= (C_1+C_2x)e^{2x}$ — 2m+1m

P.I $= \frac{1}{D^2-4D+4}\left[\frac{1+\cos 2x}{2} + 3^x\right]$ → 2m

$= \frac{1}{8} + \frac{x^2\cos 2x}{1} + \frac{3^x}{\ }$

SCHEME AND SOLUTION April-2023

UG

COURSE CODE: **COURSE:**

Question No		Marks
9 a)	$(D^2-4D+4)y = \cos^2 x + 3^x$	
	AE: $m^2-4m+4=0 \Rightarrow (m-2)^2=0 \Rightarrow m=2,2$	2
	CF: $y = (C_1+C_2x)e^{2x}$	
	PI: $y_p = \frac{1}{(D-2)^2}\left[\frac{1+\cos 2x}{2}\right] + \frac{1}{(D-2)^2} e^{x\log 3}$	
	$= \frac{1}{2}\left[\frac{1}{(D-2)^2}e^{0x} + \frac{1}{D^2-4D+4}\cos 2x\right] + \frac{3^x}{(\log 3-2)^2}$	
	$= \frac{1}{2}\left[\frac{1}{4} - \frac{1}{4D}\cos 2x\right] + \frac{3^x}{(\log 3-2)^2}$	
	$y_p = \frac{1}{2}\left[\frac{1}{4} - \frac{\sin 2x}{8}\right] + \frac{3^x}{(\log 3-2)^2}$	5
	Complete solution = CF + PI	1
b)	$x^3y''' + 3x^2y'' + 2xy' = x + \log_e x$	
	Put $x = e^z$, $x\,Dy = Dy$, $x^2\frac{d^2y}{dx^2} = D(D-1)y$	1
	$x^3\frac{d^3y}{dx^3} = D(D-1)(D-2)y$ where $D = d/dz$	
	$\therefore [D(D^2-3D+2) + 3(D^2-D) + 2D]y = e^z + z$	
	$[D^3 - 3D^2 + 2D + 3D^2 - 3D + 2D]y = e^z + z$	2
	$(D^3 - D)y = e^z + z$	
	$D(D^2-1)y = e^z + z$	
	AE: $m(m^2-1)=0$	
	$\Rightarrow m=0, m=\pm 1$	
	CF: $y = C_1 + C_2e^z + C_3e^{-z}$	
	PI $= \frac{1}{D^3-D}e^z + \frac{1}{D^3-D}z$	2+3
	$= \frac{z}{3D^2-1}e^z + \frac{1}{(-D)[1-D^2]}z$	
	$= \frac{z}{2}e^z - \frac{1}{D}[1-D^2]^{-1}(z)$	
	$= \frac{z}{2}e^z - \frac{1}{D}[z] = \frac{ze^z}{2} - \frac{z^2}{2}$	
	Complete solution = CF + P.I	

SCHEME AND SOLUTION April-2023

UG

COURSE CODE: 22MAI1B **COURSE:** FUNDAMENTALS OF LA, CALCULUS & D.E

Question No		Marks
10	a. $(m-1)^2 = 0 \Rightarrow m = 1, 1$	2m
	$C.F = (C_1 + C_2 x)e^x$	
	$y_p = Ay_1 + By_2$ where $y_1 = e^x$, $y_2 = xe^x$	1m
	$W = \begin{vmatrix} y_1 & y_2 \\ y_1' & y_2' \end{vmatrix} = e^{2x}$	1m
	$A = -\int \frac{y_2 x}{W} dx = \frac{x^2}{2}\log - \frac{x^2}{2}$	2m
	$B = \int \frac{y_1 x}{W} dx = x\log x - x$	1m
	$\therefore y = (C_1 + C_2 x)e^x + \frac{1}{4}x^2 e^x (2\log x - 3)$.	1m
	b) Given $\frac{1}{4}\frac{d^2q}{dt^2} + 10\frac{dq}{dt} + 1000q = 0$	
	$\frac{d^2q}{dt^2} + 40\frac{dq}{dt} + 4000q = 0$	2m
	$m^2 + 40m + 4000 = 0 \Rightarrow m = -20 \pm 60i$	2m
	$q = e^{-20t}[C_1 \cos(60t) + C_2 \sin(60t)]$	
	Using given condition $C_1 = q_0$, $C_2 = q_0/3$	2m
	$\therefore q(t) = e^{-20t}\left[q_0 \cos(60t) + \frac{q_0}{3}\sin(60t)\right]$	2m